ECE 310 : Wireless Communications Project

Milestone 2 (M2) Report : Research Project

How well does Wi-Fi CSI gain and phase correction hold up as bandwidth and synchronization error change? Reproducing and stress-testing Ratnam et al. (Samsung Research America, IEEE TWC 2024)

Group 4 | Monsoon Semester 2026 | Milestone 2: Technical Artifact

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1. Where we right now are after the M1 submission?

In M1 we picked Wi-Fi sensing and inside that to be particular “CSI signal pre-processing”, as our area of research. Our first plan was to stress-test the BVP motion indicator from Widar 3.0 [1] but after that our mentor asked us to re-check whether Widar 3.0 was really the right base paper as it was from 2019, its extraction code has known problems and it is not tied to an industry research lab, Hence we went back through the literature and changed our base paper to below:

V. V. Ratnam, H. Chen, H.-H. Chang, A. Sehgal and J. Zhang, "Optimal preprocessing of WiFi CSI for sensing applications," IEEE Transactions on Wireless Communications, vol. 23, no. 9, pp. 10820–10833, 2024 [2].

We chose it for multiple reasons. Firstly, It is peer reviewed and all authors are from Samsung Research America's Standards and Mobility Innovation Lab. Secondly, It deals directly with the hardware errors we care about (receiver gain error and timing/carrier-phase synchronization error in CSI). Thirdly, It gives complete mathematical models and algorithms so we can implement it ourselves even though no code is released and Finally it fits the course material well mainly OFDM, synchronization, SNR and channel estimation. Also to note that Widar 3.0 and BVP are not dropped completely instead they become the downstream check we plan for M3.

For M2 we did the following things:

(a) wrote the literature synthesis and the technical gap

(b) formalized the problem

(c) built a simulator and implemented the paper's correction algorithms together with the baselines it compares against

(d) reproduced the paper's simulation experiments as far as we could

(e) ran our own first stress tests on bandwidth and timing error.

Also, everything is present in the repository folder M2_g4-research-wifi-sensing/.

2. Literature synthesis

So, instead of summarizing papers one by one we have grouped the work by how it deals with the two errors that corrupt CSI. The table below lists all the groups and the citations in brackets refer to the reference list.

Approach	Main Idea of the approach	Who / where	What we have assumed	Weakness of the approach
Use error-robust metrics	Avoid the corrupted quantity: phase-only or amplitude-only features, CSI ratio or phase difference between two antennas	Surveyed in [2] (many CSI-sensing papers)	Two receive antennas whose phase errors are correlated, or a task that needs only amplitude or only phase	Throws away part of the CSI; cannot rebuild the true channel; tied to one task
Smooth or filter	Hampel, Savitzky-Golay, wavelet or median filters on CSI amplitude	Surveyed in [2]	Gain error looks like outliers or noise	Distorts the CSI; no model of the error
Estimate and remove the error (heuristic)	sqrt of CSI power as gain estimate; DBSCAN clustering on power levels; least-squares line fit on unwrapped phase; 802.11az timing estimator	Baselines in [2]	Whole power variation is gain, or phase is linear in sub carrier index	Not derived from a model of the hardware; fails when the true channel also changes
Estimate and remove the error (model-based)	Model gain as slow large-scale part plus discrete AGC steps; model timing and	Ratnam et al. [2] (Samsung)	Single antenna, mostly static channel (gamma close to 1), errors	Validated at 20 MHz only, on one respiration task and two receivers;

	common phase error; derive ML estimators and cheaper weighted-least-squares forms		independent between frames	no code released
Compare receivers directly	Same known signal fed to 6 chip sets; shows AGC and per-sub carrier non-linearity dominate cross-device differences; simple gain normalization recovers up to 75% of lost HAR accuracy	Portner et al. 2026 [3] (arXiv preprint, not yet peer reviewed as far as we could confirm)	Single antenna, 20 MHz, fixed hardware set	Does not look at wider bandwidth or timing error
Fix it above the signal level	Domain-invariant features (BVP/DFS) and test-time adaptation	Widar 3.0 [1]; WiTTA-Bench [4]; survey [5]	Indicators and models are not themselves affected by preprocessing quality	Never test what happens when the CSI underneath is imperfect or comes from wider-band hardware

What we take away from this, is that the field has two layers that rarely talk to each other which are the bottom layer ([2], [3]) cleans raw CSI and was tested at 20 MHz and in simple settings and other is the top layer ([1], [4], [5]) which builds generalization methods that quietly assume a clean input. The new IEEE 802.11bf amendment [6] makes sensing over wider bandwidths and several bands normal and the multi band work in [7] says explicitly that keeping access points synchronized is still unsolved and that is exactly the situation where the bottom layer's assumptions need checking for how true it is.

3. Technical gap

The gap we found is a specific one and we separate what the literature states from what is still our own hypothesis.

- Literature facts that the correction algorithms in [2] were derived and evaluated for a 20 MHz channel (64 or 256 sub carriers in simulation) with timing error limited to a few multiples of the sampling interval and their real-world test used 20 MHz and two receivers.
- Literature fact is that [2] states that when the sensing signal is very weak (gamma above about 0.95) simply ignoring power variation works best, so the benefit of fancy gain correction depends on the operating point.
- Literature also fact that in [6] and [7] describe wider-bandwidth, multi band sensing as the direction of the standard and [7] lists clock asynchrony between access points as unresolved.
- Gap which we have found (in our reading of the literature)is that, we did not find a study of how the quality of these model-based corrections changes as bandwidth (number of sub carriers) and timing error range are scaled in a controlled way, nor of what that does to the indicators built on top (BVP/DFS).
- Our hypothesis which is not yet tested is that, the gain and phase corrections that work at 20 MHz will behave differently at 80 or 160 MHz and some of the older baselines will might break first.

4. Problem formulation

We follow the model of [2]. For frame p and sub carrier k, the CSI the receiver reports is

$$\tilde{h}[p,k] = g[p] \cdot h[p,k] \cdot \exp(-j2\pi f_k \tau[p]) \cdot \exp(-j\psi[p]), \quad f_k = k / T_s$$

(Eq. 2 of [2])

where h[p,k] is the true channel we want, g[p] is the receiver gain error (a slow large-scale part plus a discrete AGC part), $\tau[p]$ is the timing error from imperfect symbol-start detection and $\psi[p]$ is the common phase error left after carrier-frequency-offset compensation. The true channel is a static part plus a dynamic part, h[p,k] = b[k] + d[p,k], where d carries the motion we want to sense. γ is the fraction of channel power in the static part.

Task: given only $\tilde{h}$, estimate g, τ and ψ for every frame so that the corrected CSI $\hat{h}[p,k] = \tilde{h}[p,k] \cdot \exp(j2\pi f_k \hat{\tau}[p]) \cdot \exp(j\hat{\psi}[p]) / \hat{g}[p]$ (Equation 3 of [2]) is as close as possible to the true channel.

For Quality measurement we use the correlation χ between the dynamic part of the corrected CSI and the true dynamic part d (Equation 29 of [2]), and report SNR = $\chi / (1 - \chi)$ in dB, as the paper does in spirit. (In [2], χ is a squared correlation. The paper does not state the exact SNR scale, so we compare ordering and trends with it, not absolute dB values.)

Research questions for the project:

- RQ1 : Do the model-based corrections of [2] beat the baselines in our own implementation, as the paper reports?
- RQ2 : How does correction quality change when the number of sub carriers grows from 64 to 512 (20 MHz to 160 MHz at 312.5 kHz spacing)?
- RQ3 : How does it change when the timing error range grows far beyond the paper's 100ns?
- RQ4 : Do the answers to RQ2 and RQ3 carry over to real receiver data and to BVP/DFS computed after correction?

Note RQ4 is planned for M3. !!

5. Proposed system and methodology

The pipeline follows the order used in [2] we first estimate and remove the gain error then estimate and remove timing and common phase error.

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Raw CSI → gain correction → phase (timing + CPE) correction → corrected CSI → quality metric χ / SNR
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5.1 Methods implemented

Step	Method	Role	Where in [2]
Gain	Eq. 6: divide by sqrt of mean CSI power	Baseline	Eq. 6
Gain	Algorithm 1: DBSCAN on power increments, then 0.1 Hz low-pass for the large-scale part	Proposed (simple)	Sec. III-C
Gain	Algorithm 3: ML estimate for a uniform AGC grid, line search over the unknown step size	Proposed (best in paper)	Sec. III-D
Phase	Eq. 21: least-squares line fit on unwrapped phase	Baseline	Eq. 21
Phase	Eq. 22: 802.11az timing estimator plus phase estimate	Baseline	Eq. 22a/b
Phase	Algorithm 4 with Eq. 25: estimate static channel, then weighted least squares with robust unwrapping	Proposed (best trade-off in paper)	Sec. IV-C, Eq. 23–26

5.2 Simulator

We wrote a CSI simulator in Python (numpy, scipy, scikit-learn). Default settings follow Sec. V-B of [2]: $T_s = 3.2$ µs, frame interval 100 ms, 300 frames per trial, $K = 64$ sub carriers, $\gamma = 0.9$, dynamic part i.i.d. complex Gaussian. Large-scale gain is a Gaussian process with 0.2 dB standard deviation below 0.1 Hz; AGC gain takes values $\{-0.5, 0, +0.5\}$ dB with probabilities $\{0.2, 0.6, 0.2\}$; timing error is uniform on [0, 100ns) and the common phase error is uniform on $[-\pi, \pi)$. Every run uses a fixed seed, so the numbers in this report can be regenerated exactly.

5.3 Experiments

As in Figures 6 and 7 of [2], to judge one correction step fairly we make the other step ideal and the gain methods are compared with perfect phase correction and phase methods with perfect gain correction. Each point below is the median over 100 random channel realizations (the paper uses 2000) and the shaded band in the plots is the inter quartile range.

- E1 (reproduction): phase methods versus γ, $K = 64$.
- E2 (reproduction): gain methods versus γ, $K = 64$.
- E3 (our stress test): phase methods versus maximum timing error, from 100 ns up to 3.2 µs (one full symbol), $K = 64$, $\gamma = 0.9$.
- E4 (our stress test): phase methods versus number of sub carriers $K = 64, 128, 256, 512$ at fixed 312.5 kHz spacing, which corresponds to 20, 40, 80 and 160 MHz, $\gamma = 0.9$.
- E5 (our stress test): gain methods versus K, same settings.

5.4 Simplifications we made (and why they matter)

- The paper's static channel uses the 802.11ax Model-C channel model where we generated the static part from a random six-tap exponential power-delay profile with 200ns spread also the frequency selectivity is roughly similar.
- We simulated only the i.i.d. dynamic component (type (i) in [2]) but not the Doppler-structured type (ii).
- 100 trials per point and not 2000 and also our graph curves are noisier and we have no significance test.

- Equation 21 in the arXiv text has a sign that is inconsistent with Equation 2 and so we fitted the negative of the unwrapped phase. Equation 16 is garbled in the arXiv text, so the Algorithm 3 uses nearest-integer rounding to the AGC grid but we still need to check both against the published PDF.
- Our simulator has no cyclic prefix or inter-symbol interference.

6. Results

6.1 E1 and E2: reproducing the paper (RQ1)

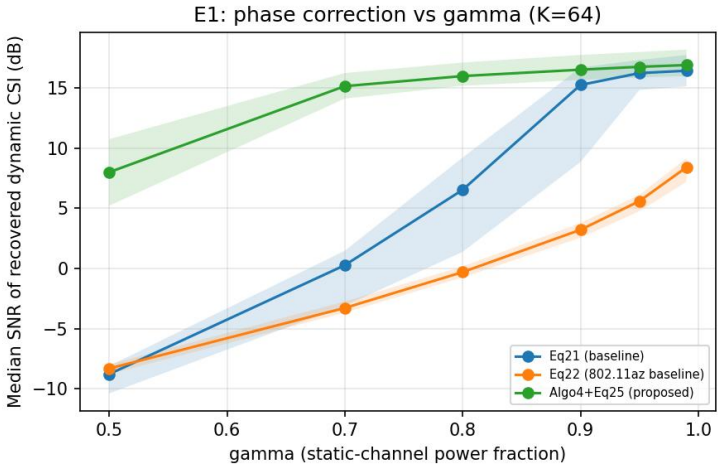


Figure 1. E1: SNR of recovered dynamic CSI after phase correction versus γ (K = 64, ideal gain correction).

γ	0.5	0.7	0.8	0.9	0.95	0.99
Equation 21 (baseline)	−8.8	0.3	6.5	15.2	16.2	16.4
Equation 22 (baseline)	−8.3	−3.3	−0.3	3.2	5.6	8.4
Algorithm 4 + Eq. 25	8.0	15.1	16.0	16.5	16.7	16.9

Median SNR in dB, 100 trials.

The ordering matches the paper: Algorithm 4 + Eq. 25 is best, Eq. 21 is second and the 802.11az estimator (Eq. 22) is worst. The proposed method is especially useful when the static part is weak (at γ = 0.5 it is about 17 dB above both baselines) yet the size of the improvement does not match the paper. At γ = 0.9 the paper reports gains of more than 1000% over both baselines for this channel type. In our runs the proposed method is only about 35% higher than Eq. 21 in linear SNR (16.5 dB against 15.2 dB), although it is about 21 times higher than Eq. 22. Above γ = 0.95 the three methods are close and the proposed one is less than 1 dB above Eq. 21. So our reproduction supports the direction of the paper's claim for phase correction but not its magnitude against Eq. 21. We think the simplified channel (Section 5.4) is a likely reason, but we have not tested that completely because there can be other possible reason for the same.

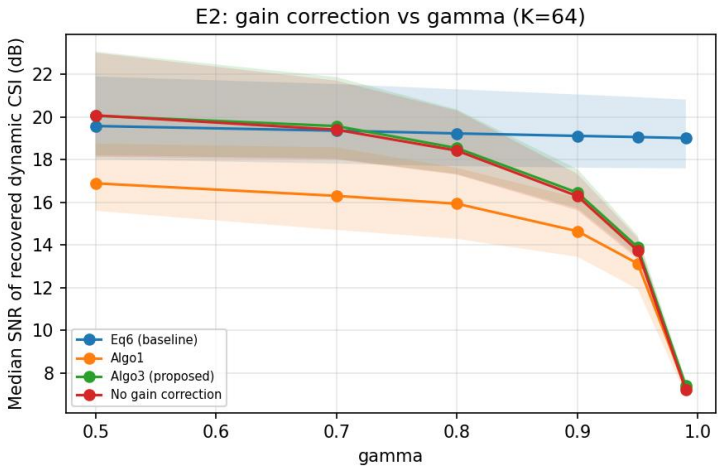


Figure 2. E2: SNR after gain correction versus γ (K = 64, ideal phase correction).

γ	0.5	0.7	0.8	0.9	0.95	0.99
Eq. 6 (baseline)	19.6	19.4	19.2	19.1	19.1	19.0
Algorithm 1	16.9	16.3	15.9	14.6	13.1	7.3
Algorithm 3	20.1	19.6	18.5	16.5	13.9	7.4
No gain correction	20.1	19.4	18.4	16.3	13.7	7.2

Median SNR in dB, 100 trials.

6.2 E4 and E5: bandwidth scaling (RQ2)

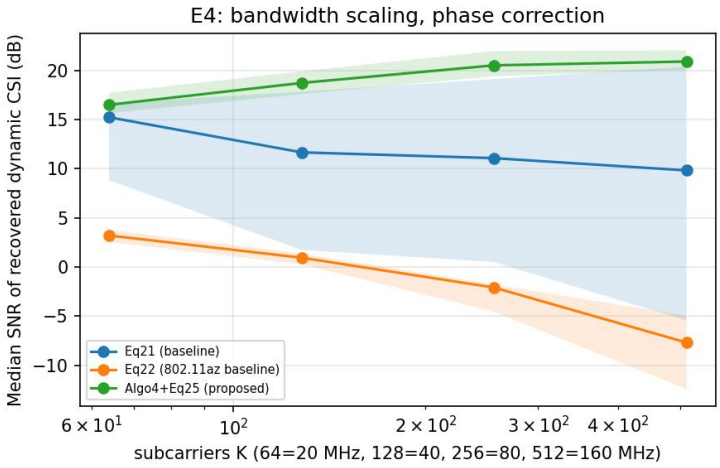


Figure 3. E4: phase correction versus number of sub carriers (20 to 160 MHz), γ = 0.9.

K (bandwidth)	64 (20 MHz)	128 (40 MHz)	256 (80 MHz)	512 (160 MHz)
Eq. 21 (baseline)	15.2	11.7	11.1	9.8
Eq. 22 (baseline)	3.2	0.9	−2.1	−7.7
Algorithm 4 + Eq. 25	16.5	18.7	20.5	20.9

Median SNR in dB, 100 trials.

This is the clearest result of M2. As bandwidth grows, the proposed phase correction improves (16.5 to 20.9 dB), while both baselines get worse (Eq. 21 from 15.2 to 9.8 dB, Eq. 22 from 3.2 to −7.7 dB). The gap between the proposed method and Eq. 21 grows from 1.3 dB at 20 MHz to 11.1 dB at 160 MHz, so a method that might look only slightly better at 20 MHz becomes much more important at the bandwidths 802.11bf-era hardware will use. The reasons for the baselines getting worse are our guesses so far and the paper itself says Eq. 22b is only accurate for a strong, flat line-of-sight channel and a wider band sees more frequency-selective fading which also makes phase unwrapping in Eq. 21 more error-prone. Here, we have not isolated either explanation as these results are from a simulated channel only.

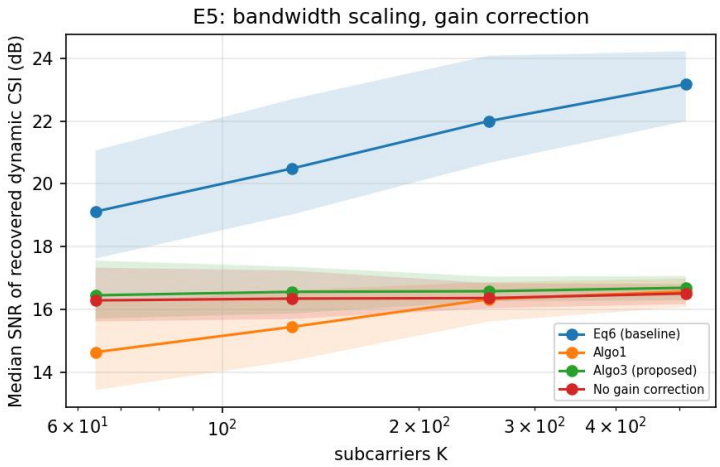


Figure 4. E5: gain correction versus number of sub carriers, γ = 0.9.

For gain correction (Figure 4) the simple baseline improves with K (19.1 to 23.2 dB), which is expected because the CSI power average has less noise with more sub carriers. Also, the Algorithm 3 stays flat at about 16.5 dB and Algorithm 1 rises only from 14.6 to 16.6 dB. But, we expected Algorithm 3 to gain from larger K since the true power fluctuation falls as $1/\sqrt{K}$ and the 0.5 dB AGC steps should become easier to see and the thing is that it does not is another sign that our Algorithm 3 may have a problem (see Section 9).

6.3 E3: timing-error range (RQ3)

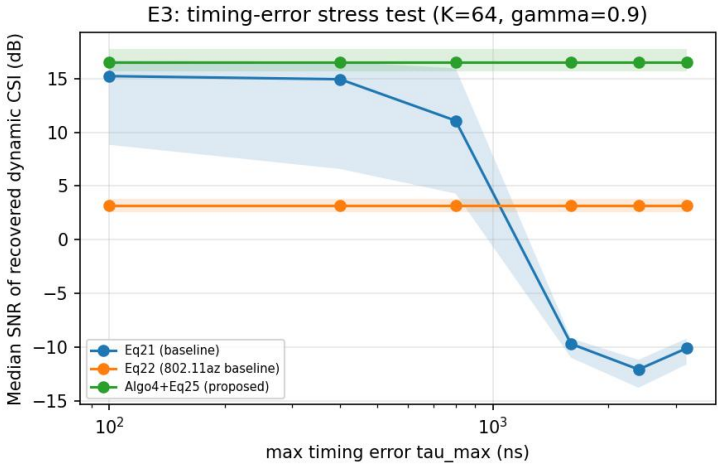


Figure 5. E3: phase correction versus maximum timing error (K = 64, γ = 0.9).

Max timing error	**100 ns**	**400 ns**	**800 ns**	**1.6 µs**	**2.4 µs**	**3.2 µs**
Eq. 21 (baseline)	15.2	14.9	11.1	−9.7	−12.1	−10.1
Eq. 22 (baseline)	3.2	3.2	3.2	3.2	3.2	3.2
Algorithm 4 + Eq. 25	16.5	16.5	16.5	16.5	16.5	16.5

Median SNR in dB, 100 trials.

Eq. 21 collapses once the timing error approaches half a symbol (1.6 µs), because it relies on unwrapping the phase and the phase step between neighbouring subcarriers then reaches ±π. The other two methods do not change at all here this looks like robustness, but we do not want to claim it as we cannot prove it. In our model the subcarrier frequencies are exactly k/T_s, so a timing error of one full symbol rotates every subcarrier by a whole number of turns and has no effect. A real receiver would also see inter-symbol interference once the error exceeds the cyclic prefix (0.8 µs), which our simulator does not include in it. So E3 only tells us that the estimators themselves are not limited by the timing range as it says nothing yet about real hardware. This needs a more realistic test in M3.

7. SOTA positioning

	Ratnam et al. 2024 [2] (Samsung) — our SOTA anchor	What we have done at M2	What we plan beyond it
Problem	Remove receiver gain error and timing/phase error from single-antenna CSI	Same problem, same model (Eq. 2)	Ask how the cleaning behaves as bandwidth and timing error scale up, and what it does to BVP/DFS
Method	Model-based ML and weighted-least-squares estimators	Implemented Algorithms 1, 3, 4 and baselines from the equations	Verify Algorithm 3; consider a targeted fix if the data shows one is needed
Evaluation	2000 simulated trials (Model-C channel), one respiration test with 75 episodes at 20 MHz	100 simulated trials with a simplified channel; no real data yet	More trials, Model-C-like channel, real multi-device data from [3]
Reproducibility	No code or data published	Code and results in our repository, seeded	Same, plus real-data scripts
Result	Proposed methods beat baselines by 40% (gain) and 200% or more (phase)	Phase: same ordering, smaller margin over Eq. 21. Gain: not reproduced	Explain or fix the gain discrepancy

How this differs from M1: at M1 our SOTA anchor was Widar 3.0 [1], which is a gesture-recognition system and not a preprocessing method. For M2 the anchor is [2], which is the preprocessing method itself, and [1] has moved to a downstream role. The closest recent related work is Portner et al. [3]. It is a preprint, so we use it as a source of public real-receiver data and not as our SOTA anchor. Industry and research-lab context: Samsung Research America [2] (this is the anchor), Huawei's Wireless Technology Lab (co-author of the survey [5], active in 802.11bf), NIST-affiliated authors of the 802.11bf evaluations [6], and the TU Delft / TU Darmstadt (SEEMOO) / University of Brescia group behind [3].

8. Limitations

- All results are from our own simulator with a simplified channel but we currently have no real CSI results yet.
- Only 100 trials per point, and no statistical test beyond that inter quartile range.
- The gain-correction results (E2, E5) disagree with the paper. Until we find out why, we do not trust our Algorithm 3 implementation.
- E3 does not model inter-symbol interference, so it cannot support any claim about real hardware.
- Single antenna only, as in [2].

9. Our plan for M3 and M4

When	Task	Why
Now to M3 (1 Nov)	Check Eq. 16, Eq. 21 and Algorithm 3 against the published PDF; re-test Algorithm 3 on K = 256 and with a more realistic channel	Settle the gain discrepancy
	Raise trials to 500 or more; add confidence intervals	Make the comparisons statistically meaningful
	Download the public multi-receiver CSI data from [3]; run baselines and our implementation of [2] on it	First real-data evidence
	Apply correction before BVP/DFS extraction on Widar 3.0 data and measure the effect on the indicator	Connect back to our M1 question (RQ4)
M3 to M4 (22 Nov)	If a clear failure appears at wide bandwidth, test a targeted fix; write final report and clean repository	Contribution beyond reproduction

10. Reproducibility :

Everything is in M2_g4-research-wifi-sensing/Code. After installing the four packages in requirements.txt, running python run_experiments.py 100 from the src folder regenerates all figures and numbers in this report (about five minutes on a normal laptop). The random seed is fixed. Code/README.md lists the simplifications from Section 5.4.

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